

## UNIT 7 Heavy Weather

The Marina is underway in the Bay of Biscay. It is about seven a.m. on Sunday morning and John Andersen is standing the four to eight watch. The weather forecast the previous evening has told of low pressure forming in the western Atlantic.

John now notices that the barometric pressure is falling rapidly. The sea is beginning to get very rough - the wind is increasing by the hour.

On the bridge the NAVTEX printer starts to work and the Chief Officer reads the message. It contains a gale warning for the Bay of Biscay.

Weather forecast for area eastern Atlantic and the Bay of Biscay:

FQNT51 LFPW 220931

NAVTEX MER698

TXT

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BAY OF BISCAY BULLETIN (METAREA 2)

METEO-FRANCE

TUESDAY 19 SEPTEMBER 2005 AT 0700 UTC.

WIND IN BEAUFORT SCALE

1 : WARNING : GALE WARNING FOR AREA BAY OF BISCAY: GALE FORCE 9  
INCREASING. STORM FORCE 10 IMMINENT

2 : GENERAL SYNOPSIS, TUESDAY 19 AT 00 UTC

LOW 990 IN BAY OF BISCAY, MOVING SOUTHEAST AND FILLING, EXPECTED  
1000 OVER SPAIN BY 22/12 UTC.

HIGH 1045 OVER SCANDINAVIA, SLOW-MOVING.

3 : FORECASTS TO WEDNESDAY 20 AT 12 UTC

IROISE, YEU :

EASTERLY 9 TO 10, DECREASING 7 TO 8 LATER. VERY ROUGH, SPRAY,  
VISIBILITY AFFECTED, SHOWERS.

ROCHEBONNE :

EASTERLY 7 TO 8, BACKING NORTHEASTERLY 6 OR 7 SOON, THEN  
DECREASING 5 TO 6 LATER. ROUGH, DECREASING TO MODERATE LATER.  
SHOWERS...

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The wind has now reached force 10. The seas are already very high with white crests and the air is white with spray. Captain James has been through many storms. He knows that he will need all his skill and experience to save the ship from damage or loss. There are many important factors to consider - the wind direction and force, the course being steered, the height of the sea, and whether the sea is short or long. In a strong headwind the vessel will pitch. When the wind is abeam the ship, we say that she is rolling. Not only the height but the length of the seas is important.

Stability is the ability of a vessel to endure high winds and seas and resist capsizing even if she heels over (or lists) heavily in the wind. Adding weight above a vessel's centre of gravity will change its stability. If the centre of gravity is raised too much, the vessel will become unstable. As a result, less tilt is needed to capsize the vessel. Removing weight from below the centre of gravity also decreases stability.

When the Marina was loaded, the Chief Mate was careful to see that the loading did not decrease the stability of the Marina. If much of the cargo weight is at the bottom of the holds, the centre of gravity is low, and the ship is said to be stiff. The breaking waves then strike the ship's hull with force and in some cases the hull might even be damaged. On the other hand, if too much cargo has been placed in the 'tween decks or on the decks, the centre of gravity is high. This makes the ship tender, and she will roll freely and rapidly. Cargo might shift and the ship might start listing. She may then be unable to recover from a roll and capsize.

Both wind and sea are now abaft the starboard beam of the Marina, which is rolling heavily in the high seas.

Captain: I think we ought to reduce speed to half speed.

Chief Officer: OK, half speed.

Captain James has reduced speed to save the Marina from being hit too violently by the waves. The storm is now becoming increasingly violent and the Marina is pounding heavily.

Captain: We shall have to heave to. We'll take her bow to the sea.  
Hand steering on!



Chief Officer: Very well, hand steering!  
Captain: Stand by Engine Room control!  
Engine Room Control: Standing by.

Outside there is only the roaring of the wind and the fierce beating of the seas, which are now mountainous, - wind, air and water. Inside the navigating bridge there is only the monotony of the orders to the Engine Room and the strained concentration of the two men.

Captain: Dead slow ahead.  
Chief Officer: Dead slow ahead.  
Captain: Hard a starboard.  
Chief Officer: Hard a starboard.  
Captain: Full ahead.  
Chief Officer: Full ahead.  
Captain: Stop her.  
Chief Officer: Stop her.

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Captain: Dead slow ahead.  
Chief Officer: Dead slow ahead.  
Captain: Is she answering the wheel?  
Chief Officer: Yes, she is.

## The Perfect Tense

### Study these sentences:

The Boatswain **has received** a number of orders because of the storm. As soon as the Third Officer **has returned** from his watch he goes to inspect the Boatswain's work.

**Have you checked** all lashings on deck and doubled them? he asks.

**Yes, I have**, answers the Boatswain.

**Have you told** the men what to expect? asks the Third Mate.

**Yes, indeed I have**.

The Boatswain **has done** a thorough job as always. He **has tightened** all the boat gripes and he **has cleared away** all loose gear from the decks. Finally he **has made sure** that all the cranes are secured.

**The Perfect Tense** expresses what someone **has done** or what **has happened**.